

Social Media

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Social media platforms for research



1 Introduction to Social Media Data

- Benefits & Limitations of Social Media Data

2 Types of Social Media Data and APIs

- Reddit
- Twitter/X
- Facebook & Instagram
- TikTok

What is Social Media Data?

- Data generated by users on social media platforms
- Includes text, images, videos, links, and metadata
- Can be used to study a wide range of topics
- Can be analyzed using a wide range of methods

Forms of social media data

- **Text**: posts, comments, messages, etc.
- **Images**: photos, memes, etc.
- **Videos**: TikToks, YouTube videos, etc.
- **Links**: URLs, hyperlinks, etc.
- **Metadata**: timestamps, geotags, etc.
- **Networks**: connections between users, groups, etc.
- **Engagement**: likes, shares, comments, etc.

Social media data collection methods

- **APIs**: Application Programming Interfaces
 - **Streaming APIs**: real-time data collection
 - **REST APIs**: historical data collection
 - Example: Reddit API
- **Web scraping**: automated data collection from websites
- **Manual data collection**: copying and pasting data
- **Hybrid methods**: combining methods
- **Data sharing**: using data collected by others

Benefits of Social Media Data

- **Large-scale:** millions of users, billions of posts
- **Real-time:** data are available as events unfold
- **Rich:** text, images, videos, links, and metadata
- **Interactive:** users can respond to each other
- **Cost-effective:** data are often free or low-cost
- **New opportunities:** can be used to study sensitive topics and/or niche communities
- Multiple forms of research:
 - **Platform focused:** studying the platform itself
 - **As a data source:** using data to answer research questions

Limitations of Social Media Data

- **Accessibility** e.g., private posts are protected; access/rules can change any time
- **Non-representativeness**: users are non-random sample of the general population; what users share is non-random
- **Reproducibility** e.g., companies may limit what researchers can make public
- **Rules and terms of service**: written to be (intentionally) ambiguous
- **Ethics**: considerations include how the data are used, (in)ability to gain consent from research subjects, and protecting research subjects from harm

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- Forum for discussion, news aggregation, and content rating
- Subreddits (communities) are both user-created and user-moderated
- Used for research across disciplines, including political science and policy
 - Some examples include:
 - political subreddits (e.g., r/politics, r/worldnews, and r/The_Donald)
 - mental health subreddits (e.g., r/depression, r/suicidewatch, r/anxiety, r/bipolarreddit)
 - drug use subreddits (e.g., r/opiates)

Benefits

- 100 million daily active users worldwide
- Mostly public, and accessible via APIs at no cost
- Can be used to target people with specific interests
- Can be used to study sensitive topics
- Can be used to study niche communities

Challenges

- **Representativeness**: skews young, male
- **Pseudonyms** are the norm; user location information often unavailable
- Posting is **strategic**; top posts get more interactions
- Additional **ethical considerations** may apply to certain populations
- Subreddits have **own norms, cultures, and moderation practices**

Reddit research examples

- Collective identity formation and the extreme right on Reddit (Gaudette, Scrivens, Davies & Frank 2020)
- COVID-19 sentiment analysis using university subreddit data (Yan & Liu 2022)
- The evolution of the Manosphere across the web (Ribeiro et al 2021)
- Polarization and political interactions on Reddit (Morales, Monti, & Starnini 2021)

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No echo in the chambers of political interactions on Reddit

[Gianmarco De Francisci Morales](#) ✉, [Corrado Monti](#) ✉ & [Michele Starnini](#) ✉

[Scientific Reports](#) **11**, Article number: 2818 (2021) | [Cite this article](#)

34k Accesses | **89** Altmetric | [Metrics](#)

- Are voters polarized?
- Reddit users who are Clinton supporters shown to be likely to respond to out-group partisans
- Contradicts previous findings of homophilous networks
- **Methodology**: collected data from 2016 election-related subreddits
- Contradicts previous research on echo chambers and filter bubbles

Benefits

- 100 million daily active users worldwide (but, declining / stagnating)
- Mostly public (if you can pay!)
- Use of real names / identifying information is more common
- Can be used to target people with specific interests
- Used for research across disciplines, including political science

Challenges

- **Representativeness**: skews young
- **Pseudonyms** are the norm; user location information often unavailable
- Posting is **strategic**; top posts get more interactions
- **Restricted accessibility** (varies by API type); access comes at a cost

Documentation

Search the docs



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X API



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- Access to Ads API
- \$200 per month
- \$175 per month with annual subscription (\$2,100 total)
- 2 Top-ups allowed

Pro

For startups scaling their business

- Rate-limited access to suite of v2 endpoints, including search and filtered stream
- 1,000,000 Posts per month - GET at the app level
- 300,000 Posts per month - posting limit at the app level
- 1 Project
- 3 Apps per Project with unique Environment (Development/ Production/ Staging)
- Login with X
- Access to Ads API
- \$5,000 per month
- \$4,500 per month with annual subscription (\$54,000 total)
- 2 Top-ups allowed

Notable papers using Twitter/X (1/2)

- **Barberá et al (2019)** Agenda-setting dynamics between legislators and the public using Twitter issue attention time series.

Barbera P, Casas A, Nagler J, et al. (2019). "Who Leads? Who Follows? Measuring Issue Attention and Agenda Setting by Legislators and the Mass Public Using Social Media Data" *American Political Science Review*.

- **Larson et al (2019)** Social networks and offline protest participation (Charlie Hebdo protest in France).

Larson, J. M., Nagler, J., Ronen, J., & Tucker, J. A. (2019). "Social networks and protest participation: Evidence from 130 million Twitter users." *American Journal of Political Science*.

- **Bail et al. (2018)** Field experiment exposing Twitter users to opposing political views; effects on polarization.

Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H., Hunzaker, M. F., ... & Volfovsky, A. (2018). "Exposure to Opposing Views on Social Media Can Increase Political Polarization." *Proceedings of the National Academy of Sciences*.

Notable papers using Twitter/X (2/2)

- **Huszar et al. (2022)** Algorithmic amplification of political content on Twitter.

Huszar, F., Ktena, S. I., O'Brien, C., Belli, L., Schlaikjer, A., & Hardt, M. (2022). "Algorithmic Amplification of Politics on Twitter." *PNAS*.

- **Flamino et al. (2023)** Polarization of news media and influencers on Twitter in the 2016 and 2020 US elections.

Flamino, J., Galeazzi, A., Feldman, S., Macy, M. W., Cross, B., Zhou, Z., ... & Szymanski, B. K. (2023). "Political Polarization of News Media and Influencers on Twitter in the 2016 and 2020 US presidential elections." *Nature Human Behaviour*.

- **Baribi-Bartov et al. (2024)** Concentration of fake news sharing among "supersharers" on Twitter.

Baribi-Bartov, S., Swire-Thompson, B., & Grinberg, N. (2024). "Supersharers of Fake News on Twitter." *Science*.

Benefits

- 3.35 billion daily active users worldwide (across all Meta apps)
- Some elements are public
- Offers other forms of curated data (e.g., ads)
- Can be used to target people with specific interests
- Some (very limited) access for researchers and journalists

Challenges

- **Representativeness**: Facebook skews older than other platforms; most accounts are private
- Meta **tightly controls** the information researchers can use from its platforms (bans scraping and has taken action against violators of ToS)

But, **Meta makes some data available**:

- Meta Content Library and API (replaced CrowdTangle, 2016-2024)
 - Social media platform data limited to public accounts / groups only
- Meta Data for Good; AI for Good
 - De-identified data products from Meta's infrastructure

Meta Data for Good

Infrastructure

Electrical Distribution Grid Maps

This predictive model can map medium-voltage electrical distribution infrastructure in any country.

[→ Learn More](#)

Inclusive Internet Index

The Inclusive Internet Index measures whether the internet is accessible and affordable, and how it affects social and economic mobility.

[→ Learn More](#)

Commuting Zones

Commuting Zones are geographic areas where people live and work and are useful for understanding local economies, as well as how they differ from traditional boundaries.

[→ Learn More](#)

Infrastructure Route Study

This Infrastructure Route Study contains a collection of open source data collected for expanding fiber routes to extend Internet connectivity in the Democratic Republic of the Congo.

[→ Learn More](#)

Global Urban Areas

The Global Urban Areas dataset is a high resolution dataset consists of 37,000 polygonal areas around the globe, and provides enough detail for most use-cases.

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Network Coverage Maps

Network Coverage Maps show where people on Facebook have cellular connectivity.

[→ Learn More](#)

Base

The Base Dataset is a foundational element of navigation, local search, and many other location-based applications.

[→ Learn More](#)

Buildings

An alpha version of open map data showcasing over 6 million buildings that have been derived from the US Geological Survey (USGS) 3D Elevation Program Light Detection and Ranging and open building footprint data.

[→ Learn More](#)

Divisions

The Overture Maps Foundation (OMF) divisions dataset provides information on official and non-official organizations of people at the country, region, province, city, and neighborhood level.

[→ Learn More](#)

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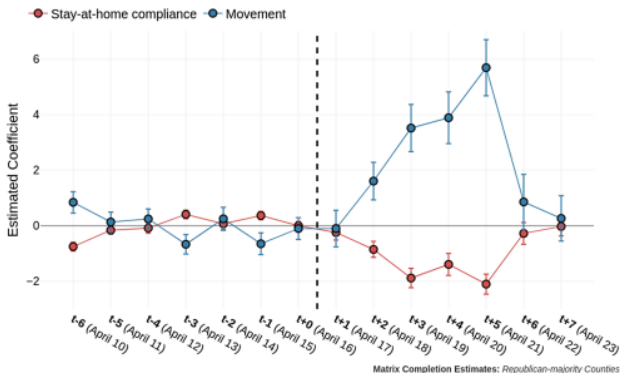
Elite Cues and Noncompliance

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Political leaders increasingly use social media to speak directly to voters, but the extent to which elite cues shape offline political behavior remains unclear. In this article, we study the effects of elite cues on noncompliant behavior, focusing on a series of controversial tweets sent by US President Donald Trump calling for the “liberation” of Minnesota, Virginia, and Michigan from state and local government COVID-19 restrictions. Leveraging the fact that Trump’s messages exclusively referred to three specific US states, we adopt a generalized difference-in-differences design relying on spatial variation to identify the causal effects of the targeted cues. Our analysis shows that the President’s messages led to an increase in movement, a decrease in adherence to stay-at-home restrictions, and an increase in arrests of white Americans for crimes related to civil disobedience and rebellion. These findings demonstrate the consequences of elite cues in polarized environments.

FIGURE 5. Dynamic Effects of “Liberate” Cues on Mobility in Republican Counties



Note: Matrix completion coefficient estimates and 95% confidence intervals for the effect of the cues on movement (blue) and stay-at-home compliance (red) in Republican-majority counties (e.g., model 3 in Tables 1 and 2). The counterfactual includes Republican-majority counties around the country that were not targeted in the President calls for liberation and were under the same mandatory state restrictions.

Meta research example



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HOME > SCIENCE > VOL. 381, NO. 6656 > HOW DO SOCIAL MEDIA FEED ALGORITHMS AFFECT ATTITUDES AND BEHAVIOR IN AN ELECTION CAMPAIGN?

🔒 | SPECIAL ISSUE RESEARCH ARTICLE | SOCIAL MEDIA



How do social media feed algorithms affect attitudes and behavior in an election campaign?

ANDREW M. GUESS , NEIL MALHOTRA , JENNIFER PAN , PABLO BARBERÁ , HUNT ALLCOTT, TAYLOR BROWN , ADRIANA CRESPO-TENORIO, DREW DIMMERY 
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, BRENDAN NYHAN , CARLOS VELASCO RIVERA, JAIME SETTLE , DANIEL ROBERT THOMAS, EMILY THORSON , REBEKAH TROMBLE , ARJUN WILKINS, MAGDALENA WOJCIESZAK , BEIXIAN XIONG, CHAD KIEWIET DE JONGE, ANNIE FRANCO, WINTER MASON , NATALIE JOMINI STROUD , AND JOSHUA A. TUCKER 

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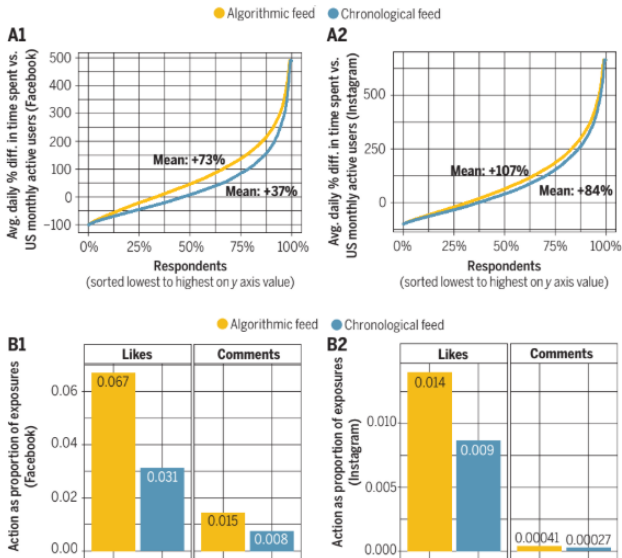


Guess, A. M., Malhotra, N., Pan, J., Barberá, P., Allcott, H., Brown, T., ... & Tucker, J. A. (2023). How do social media feed algorithms affect attitudes and behavior in an election campaign?. *Science*, 381(6656), 398-404.

How to do the **Facebook and Instagram algorithms** affect engagement and attitudes?

- **Methodology:** Working with Meta, FB and IG feeds were randomly switched to **chronological order**, instead of algorithms
- **Results:**
 - Engagement **decreased** when feeds were switched to chronological order
 - Less time spent on FB and IG; fewer likes, comments, shares
 - Altering the new feed algo did **not** significantly alter levels of issue polarization, affective polarization, and political knowledge (with some exceptions)

Chronological feeds reduce engagement



Chronological feeds increase moderate content

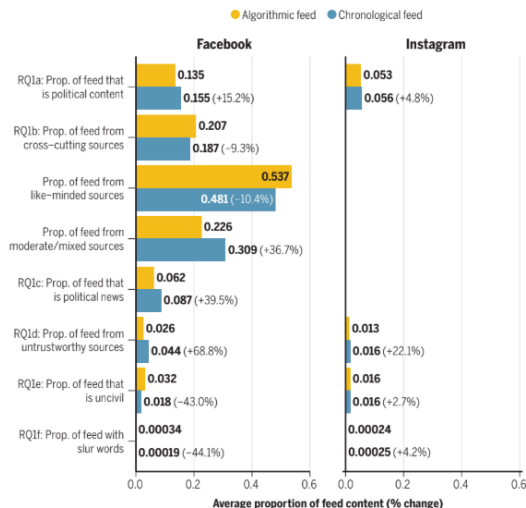


Fig. 2. Estimated changes in prevalence of feed content on both Facebook and Instagram. (Left) Facebook. (Right) Instagram. Values are average unweighted proportions within each group, with percent changes relative to the Algorithmic Feed control group in parentheses. All differences are significant at the $p < 0.005$ level, except RQ1f for Instagram ($p < 0.05$); confidence intervals are thus not shown. RQ1b and RQ1c were not tested for Instagram because political and ideology classifications are not available on that platform. Fully specified regression models with survey weights are reported in the SM, section S2.2.

Experimental: Paid Advertising on Facebook



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Descriptive example: Bond & Messing 2015

"We begin by creating a matrix in which each column represents a user and each row represents an official Facebook page about politics. We limit our data to Facebook users in the United States who are over age 18 who endorse, or "like," at least two political pages. This leaves us with approximately 6.2 million users and 1,223 pages and 18 million like actions from users to pages. An example of the first ten rows and first ten columns..."

TABLE 1. The First Ten Rows of the User by Political Page Matrix

	user 1	user 2	user 3	user 4	user 5	user 6	user 6	user 8	user 9	user 10
Barack Obama	1	1	1	.	.	.	.	.	.	.
Mitt Romney	1	.	.	.	.	.	.	1	.	.
Howard Dean	1	.	.	.	.	.	.	.	.	1
Joe Biden	.	.	.	.	.	.	.	.	.	.
Mike Bloomberg	1	.	1	.	.	.	.	.	.	.
Anthony Weiner	1	.	.	.	.	.	.	.	.	.
Deval Patrick	1	.	.	.	.	.	.	.	.	.
Diane Feinstein	1	.	1	.	.	.	.	.	1	.
Sarah Palin	.	.	.	.	.	.	.	.	.	.
Nancy Pelosi	.	.	.	.	.	.	1	.	.	.

Note: Entries in the matrix are dichotomous, where 1 means that the user has liked the page and . means that the user has not.

Bond, R., & Messing, S. (2015). Quantifying social media's political space: Estimating ideology from publicly revealed preferences on Facebook. *American Political Science Review*, 109(1), 62-78.

Benefits

- Over 1 billion daily active users
- Rich data (users, engagement, video transcripts, etc.)
- Some API access for researchers in the US and Europe

Challenges

- **Controversial(?)**: Concern about user data privacy / national security
 - but note that [Facebook collects even more data](#) about you!
- User base skews [young](#) (18-34), getting older over time
- Most users are viewing, not posting, videos
- TikTok Research API may not provide access to full content of platform ([See systematic audit by Pearson et al 2025](#))

TikTok Research Example: Munger & Li 2025



Figure 1. Screenshots from treatment videos. (a) Biden badass. (b) Biden thirst trap. (c) Trump badass. (d) Trump thirst trap.

Munger, K., & Li, V. (2025). Thirst traps and quick cuts: The effects of TikTok "edits" on evaluations of politicians. *Social Media+ Society*, 11(2), 20563051251329990.